THE UNIVERSITY OF HONG KONG

DEPARTMENT OF MATHEMATICS

MATH2014: Multivariable Calculus and Linear Algebra

May 18, 2017 TIME: 2:30p.m.-5:00p.m.

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Answer ALL TEN questions.

Notes:

- You should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully before you write.**
- You must start each question on a new page. You should write down the question number on the top right hand corner of each page.
- If your answer to a question spans over more than one page, you must indicate clearly on each page (except the last) that the answer will continue on the subsequent page.
- If you do not otherwise specify, you are assumed to be answering the questions in order. In particular, indicate on each question where your answer to the next question is if it does not follow immediately. (For example you may write 'turn back 3 pages for Question 5' at the end of Question 4.)
- Whenever there is a blank page it will be assumed that all subsequent pages are also blank. If this is not true, indicate clearly on the page which you would otherwise leave blank.

1. (10 marks) Find and classify all the critical points for $f(x, y) = 3x^2y + y^3 - 3x^2 - 3y^2 + 2$.

2. (10 marks) Suppose you are running a company, producing some sort of widget that requires steel as a raw material. Your costs consist of human labor, which is \$20 per hour, and the steel itself, which runs for \$170 per ton. Suppose your revenue is modeled by the following equation:

$$R(h, s) = 200h^{\frac{2}{3}}s^{\frac{1}{3}},$$

where h represents hours of labor and s represents tons of steel. If your budget is \$20000, what is the maximum possible revenue? (Round your answers to the nearest dollar.)

3. (10 marks) Let $f(x, y) = 2x^2y^2 + x^2y - 2x - y^2$. Using ONE iteration of Newton-Raphson's method, estimate one critical point of f with the initial guess $(1, -1)$. (Round your answers to 4 decimal places.)

4. (10 marks) Evaluate $\int\int_R \left(\frac{x-y}{x+y+2}\right)^2 dxdy$ over the region R below.

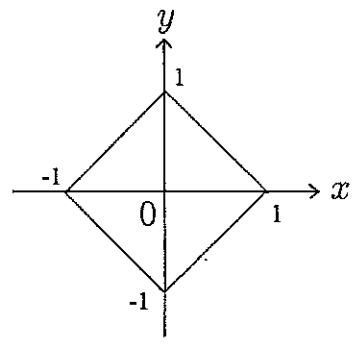

Figure 1: Region R in question 4

5. (10 marks) Find the volume of the solid that lies above the cone $z = \sqrt{x^2 + y^2}$ and below $x^2 + y^2 + z^2 = z$.

6. (10 marks) Compute $\int_{-1}^{1}\int_{0}^{\sqrt{1-x^2}}\int_{(x^2+y^2)^3}^{5} \sqrt{x^2+y^2}\,dzdydx$.

7. (10 marks) Find the area of the region bounded by the curves $r = \cos\theta$ and $r = \sin\theta$.

8. (10 marks) Solve the system of linear equations

$$\begin{cases} 2x_1 - 2x_2 + 3x_3 = 10 \\ 3x_1 - 3x_2 + 2x_3 = 5 \\ 4x_1 + 2x_2 - 5x_3 = -6 \end{cases}$$

by

(a) Inverse Matrix method.

(b) Cramer's rule.

9. (10 marks) Given that

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -2 & 3 & -2 \\ -1 & 1 & 0 \end{bmatrix}$$

(a) Find all the eigenvalues and associated eigenvectors of A (Let the first non-zero entry of each eigenvector to be 1).

(b) Find a non-singular matrix P such that $P^{-1}AP$ is a diagonal matrix.

(c) Hence, evaluate A^{2017}.

10. (10 marks) Read the following information and answer the question.

A cryptogram is a message written according to a secret code (the Greek word kryptos means hidden). This question describes a method of using matrix multiplication to encode and decode messages.

Begin by assigning a number to each letter in the alphabet (with 0 assigned to a blank space), as follows.

$0 = _$	$1 = A$	$2 = B$	$3 = C$	$4 = D$	$5 = E$
$6 = F$	$7 = G$	$8 = H$	$9 = I$	$10 = J$	$11 = K$
$12 = L$	$13 = M$	$14 = N$	$15 = O$	$16 = P$	$17 = Q$
$18 = R$	$19 = S$	$20 = T$	$21 = U$	$22 = V$	$23 = W$
$24 = X$	$25 = Y$	$26 = Z$			

Then the message is converted to numbers and partitioned into uncoded row matrices, each having n entries.

Example 1: Write the uncoded row matrices of size 1×3 for the message *MEET ME MONDAY.*

Partitioning the message (including blank spaces, but ignoring punctuation) into groups of three produces the following uncoded row matrices.

$$\begin{array}{cccccccccccccccc} [13 & 5 & 5] & [20 & 0 & 13] & [5 & 0 & 13] & [15 & 14 & 4] & [1 & 25 & 0] \\ M & E & E & T & __ & M & E & __ & M & O & N & D & A & Y & __ \end{array}$$

Note that a blank space is used to fill out the last uncoded row matrix.

Example 2: Use the matrix
$$A = \begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$$

to encode the message MEET ME MONDAY.

The coded row matrices are obtained by multiplying each of the uncoded row matrices found in Example 1 by the matrix A, as follows.

Uncoded Row Matrix	*Encoding Matrix A*		*Coded Row Matrix*
$[13 \quad 5 \quad 5]$	$\begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$	$=$	$[13 \quad -26 \quad 21]$
$[20 \quad 0 \quad 13]$	$\begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$	$=$	$[33 \quad -53 \quad -12]$
$[5 \quad 0 \quad 13]$	$\begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$	$=$	$[18 \quad -23 \quad -42]$
$[15 \quad 14 \quad 4]$	$\begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$	$=$	$[5 \quad -20 \quad 56]$
$[1 \quad 25 \quad 0]$	$\begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$	$=$	$[-24 \quad 23 \quad 77]$

The sequence of coded row matrices is

$$[13 \quad -26 \quad 21][33 \quad -53 \quad -12][18 \quad -23 \quad -42][5 \quad -20 \quad 56][-24 \quad 23 \quad 77].$$

Finally, removing the brackets produces the cryptogram below.

$$13 \quad -26 \quad 21 \quad 33 \quad -53 \quad -12 \quad 18 \quad -23 \quad -42 \quad 5 \quad -20 \quad 56 \quad -24 \quad 23 \quad 77$$

On the other hand, we can use the inverse of A to decoding the cryptogram.

Question: Please use the matrix:

$$A = \begin{bmatrix} 3 & -4 & 2 \\ 0 & 2 & 1 \\ 4 & -5 & 3 \end{bmatrix}$$

to decoding the cryptogram: 112 -140 83 19 -25 13 72 -76 61 95 -118 71 20 21 38 35 -23 36 42 -48 32.

******* **END OF PAPER** *******